Mid-term Exam Review

Science & Technology Innovation Management

Lecturer: Dr. Do Thi Hai | USTH, Master Programme

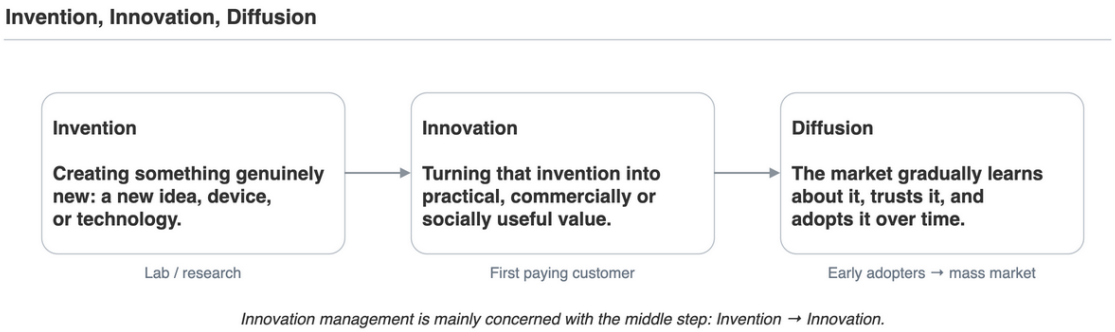
Exam: 23/10/2026, 14:55–15:55 (written, 1 hour)

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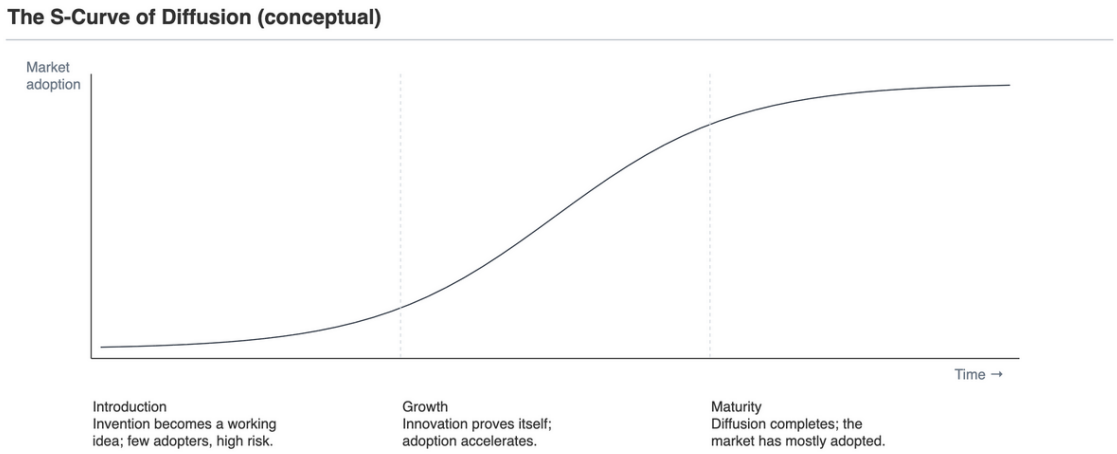
1. Definition of innovation

"Definition of innovation."

Innovation is the successful exploitation of new ideas. It is the full journey from an idea to real, delivered value, not just the moment a new idea appears. Three terms are easy to confuse and worth separating clearly. **Invention** is the creation of something genuinely new, a new idea, device, or technology, often born in a lab or a research project. **Innovation** is what happens next: taking that invention and turning it into something practical that customers will actually buy, or that solves a real social problem, so it generates commercial or social value. An invention that never reaches a market or a user is not yet an innovation. **Diffusion** is the stage after that: the gradual process by which the wider market learns about the innovation, trusts it, and adopts it, spreading from a small group of early users to the mass market over time. Innovation management, as a discipline, is concerned mainly with the middle step: deliberately organizing people, resources, and decisions so that promising ideas survive the journey from invention to adopted, value-creating innovation.



Innovation management is mainly concerned with the middle step: Invention → Innovation.



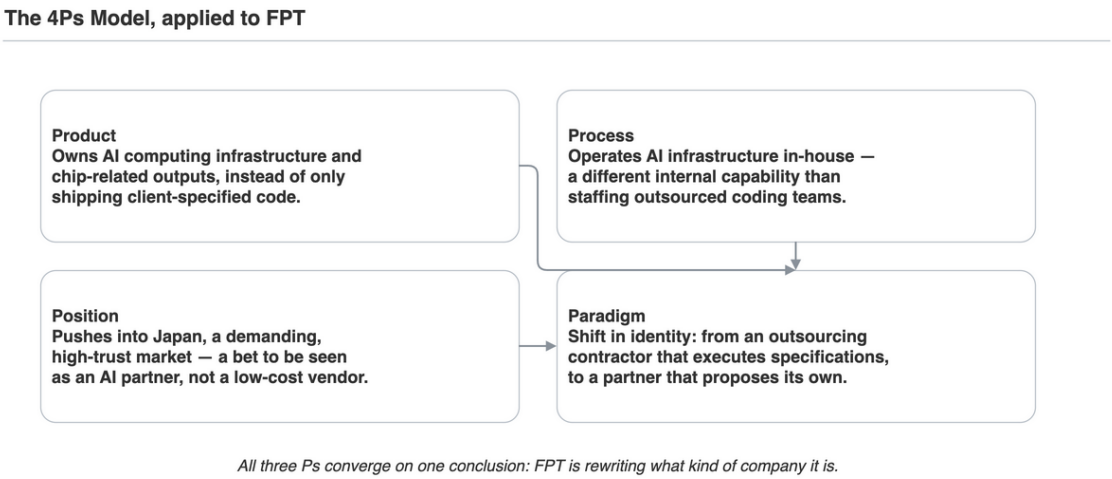
2. The “4Ps” model (Tidd & Bessant)

“What does the “4Ps of Innovation” model by Tidd & Bessant entail? Provide an example of how the innovation strategies of Product, Process, Position, and Paradigm are applied at a major technology company (such as Apple, Viettel, or FPT).”

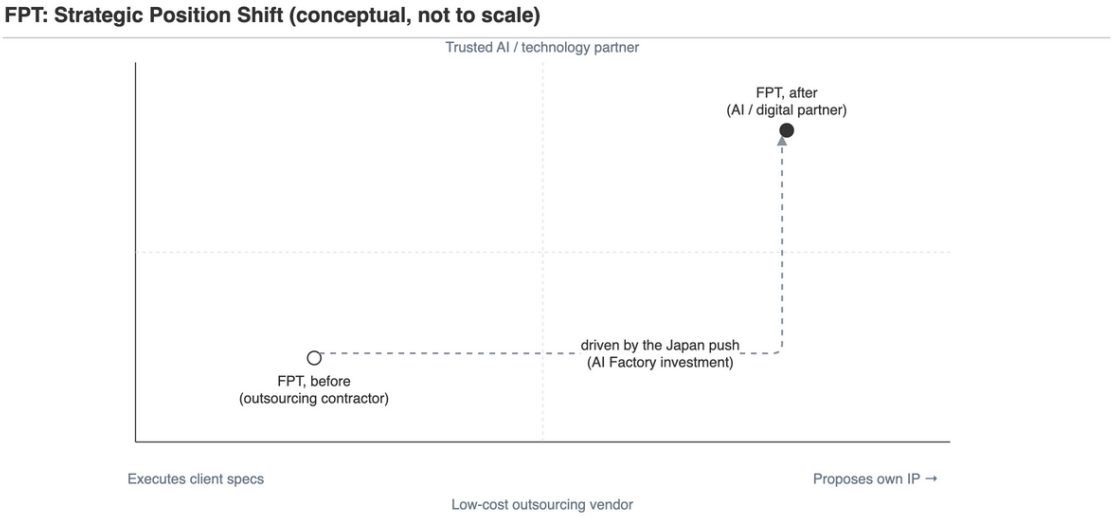
Tidd and Bessant argue that innovation is not limited to inventing a new gadget; a company can innovate along four different dimensions, often at the same time. **Product** innovation changes what the organization offers. **Process** innovation changes how it is made and delivered. **Position** innovation changes who it is sold to and how it is positioned against competitors. **Paradigm** innovation is the deepest and rarest kind: a change in the underlying business model or in what the organization fundamentally is.

FPT is the sharpest single case because all four Ps point in the same strategic direction at once: away from being a pair of hands for hire, toward being a partner with its own intellectual property. Its Product innovation is the clearest break with its own past: instead of only shipping software a foreign client specifies, FPT now builds and owns AI computing infrastructure and chip-related offerings it decided on its own were worth building. Its Process innovation follows directly: operating AI infrastructure in-house demands a different internal capability than staffing outsourced coding teams. Its Position innovation is deliberate and risky: pushing into Japan, a demanding, high-trust market, is a bet to be seen as a serious AI partner rather than a low-cost vendor. The Paradigm shift is the conclusion the other three Ps drive toward: FPT is rewriting what kind of company it is, from a contractor that

executes someone else's specification to a partner that proposes its own.



All three Ps converge on one conclusion: FPT is rewriting what kind of company it is.



3. Risks in the "Implementing" phase

"During the "Implementing" phase, what risks typically arise when transforming a technical concept into a commercial product or service? How can one manage both technological and market risks?"

The Implementing phase is where an idea that survived earlier screening is finally committed to real resources. Uncertainty is still high but the cost of being wrong rises sharply, which is why this phase concentrates two distinct families of risk. **Technological risk** is the risk that the idea does not hold up outside a controlled setting: it may not scale, may cost far more to manufacture than expected, or may behave unpredictably in real conditions the lab never tested for. **Market risk** is separate and equally dangerous: even a technically sound product may fail commercially, because customers do not want it as assumed, demand is smaller than

projected, or a rival reaches the same customers first.

Managing both means refusing to commit all resources at once and instead learning in stages. **Stage-gate** reviews let a weak idea be stopped early rather than discovered weak only after full launch. A **minimum viable product** tests whether the core technical idea works, cheaply, before polishing it. A **small pilot** with real customers tests demand directly. Critically, the two tracks should run **in parallel, not sequentially**: a product that is technically perfect but commercially unwanted is just as much a failure as one that is desired but technically broken.

Risk in the Implementing Phase

Technological risk

The concept may not scale, may be too costly to manufacture, or may not work reliably outside the lab.

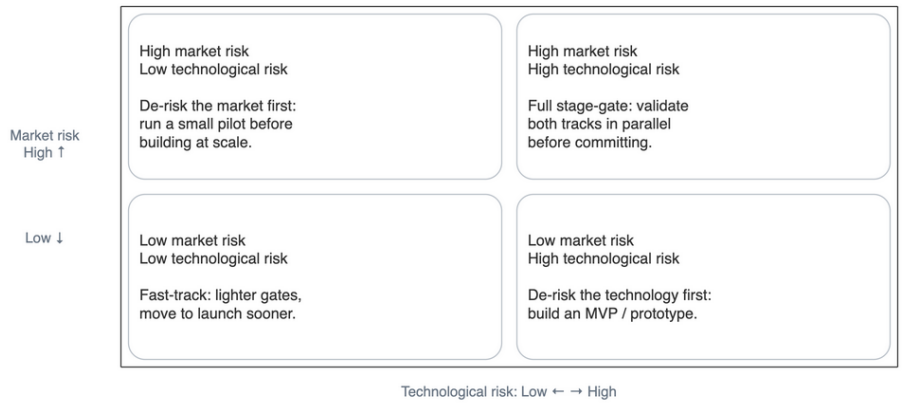
Market risk

Customers may not want it, demand may be smaller than expected, or a competitor may reach them first.

Managed together, in parallel — neither risk is tested before the other:

- Stage-gate reviews — kill or fund the project at each checkpoint.
- MVP / prototype — test technical feasibility cheaply before polishing.
- Small pilot — test real demand with real customers before scaling.

Risk Matrix: Implementing Phase (conceptual)



4. The role of innovation management (5 official roles)

"Analyze the role of innovation management for businesses. Provide illustrative examples."

Innovation management is not simply the act of having good ideas; it is the discipline of organizing a business so that good ideas reliably turn into results, through five roles: **(1) creating and sustaining competitive advantage**, making an edge repeatable rather than a one-time success; **(2) integrating technology,**

market, and organization, so a brilliant solution and a real opportunity do not pull in different directions; **(3) transforming ideas into real value**, the harder work of shipping past the idea stage; **(4) building and optimizing dynamic capabilities**, the organizational muscle to adapt as conditions shift; and **(5) managing risk and optimizing resources**, expanding responsibly without exposing the core business.

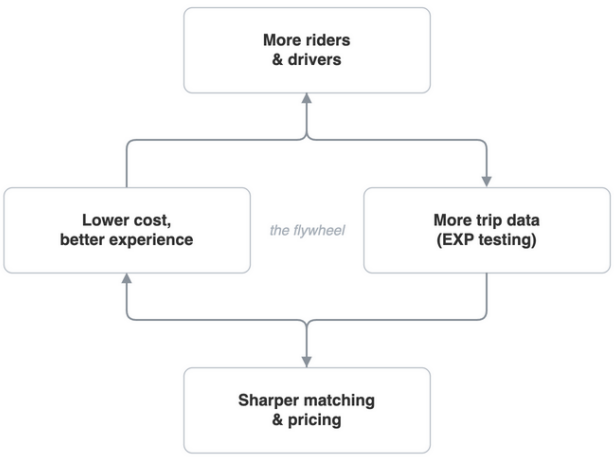
Grab shows all five at once. Its continuous, data-driven experimentation in matching and pricing sustains advantage rather than relying on one breakthrough. Its super-app strategy, bundling rides, delivery, and payments into one platform, integrates technology, market reach, and organization into a single loop. Its map and location-data tool, built first to solve an internal problem before being sold externally, turns an internal idea into real value. Its disciplined, gradual cut in promotional spending as the business matured shows dynamic capability. Its dedicated fraud-defense system, and the years spent securing financial licenses before offering lending, show risk managed deliberately while still expanding into new lines.

Five Roles of Innovation Management, Grab

1. Competitive advantage	2. Integration	3. Ideas → value	4. Dynamic capabilities	5. Risk & resources
Continuous A/B testing keeps matching and pricing sharp.	Super-app: rides, delivery and payments fused into one loop.	Internal map tool later sold as a B2B product.	Disciplined cut in promo spend as the business matured.	Fraud-defense system; licenses secured before launching lending.

Each role is a lens on the same company, not five separate companies.

Grab's Reinforcing Flywheel (conceptual, Roles 1 & 2)



Each turn of the loop makes the next turn cheaper and faster to win — sustaining competitive advantage.

Good luck this afternoon, 14:55–15:55!